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ER

ER{M,D} - a visual summary

Entity Relationship Model (ERM)

- Basis of an entity relationship diagram (ERD)
- ERD depicts the:
 - Conceptual database as viewed by end user
 - Database's main components
 - Entities
 - Attributes
 - Relationships
- Entity - Refers to the entity set and not to a single entity occurrence

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ATTRIBUTES - one of the MOST important concepts [EVER]!

Attributes

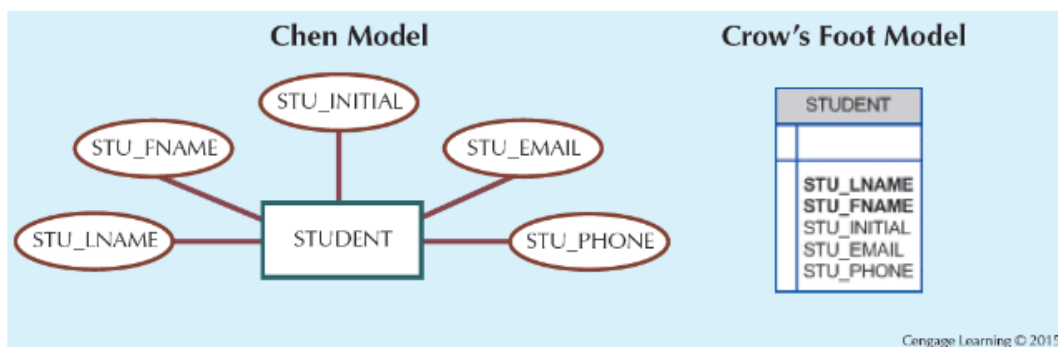
- Characteristics of entities
- **Required attribute:** Must have a value, cannot be left empty
- **Optional attribute:** Does not require a value, can be left empty
- Domain - Set of possible values for a given attribute
- **Identifiers:** One or more attributes that uniquely identify each entity instance

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An identifier is also called a KEY, or PRIMARY KEY - this is one of the 'key' concepts in all of database theory!! We'll talk much more about keys later.

Figure 4.1 - The Attributes of the Student Entity: Chen and Crow's Foot



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Attributes - so many kinds

Attributes

- **Composite identifier:** Primary key composed of more than one attribute
- ~~Compound~~ **Composite attribute:** Attribute that can be subdivided to yield additional attributes
- **Simple attribute:** Attribute that cannot be subdivided
- **Single-valued attribute:** Attribute that has only a single value
- **Multivalued attributes:** Attributes that have many values

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FYI - here is a page on the various types of attributes.

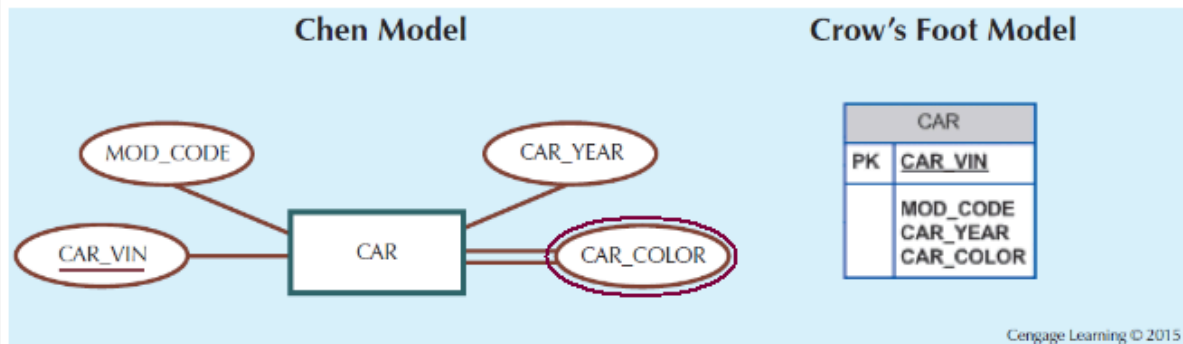
Attributes

- **Multivalued attributes:** Attributes that have many values and require creating:
 - Several new attributes, one for each component of the original multivalued attribute
 - A new entity composed of the original multivalued attribute's components
- **Derived attribute:** Attribute whose value is calculated from other attributes
 - Derived using an algorithm

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Figure 4.3 - A Multivalued Attribute in an Entity



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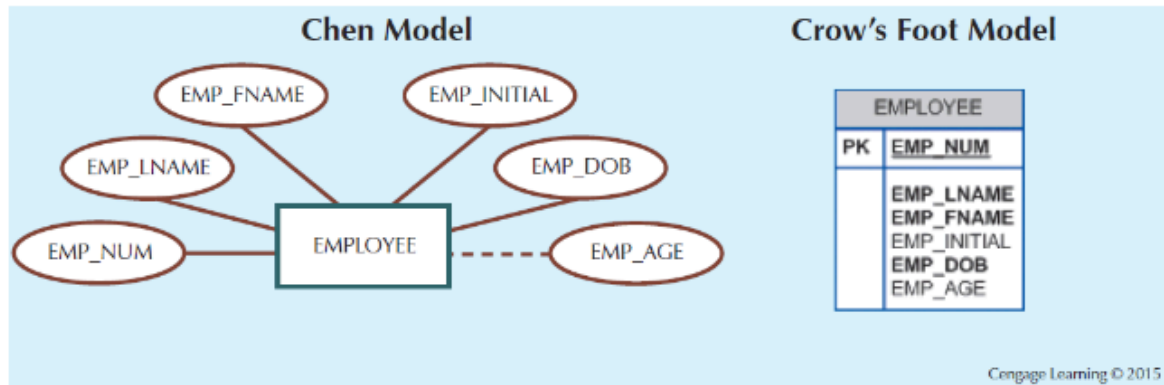
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In Crow's Foot notation, 'bold' attributes are 'required' (can't be null).

Derived attributes

Figure 4.6 - Depiction of a Derived Attribute



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Table 4.2 - Advantages and Disadvantages of Storing Derived Attributes

	STORED	NOT STORED
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

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RELATIONSHIPS between entities

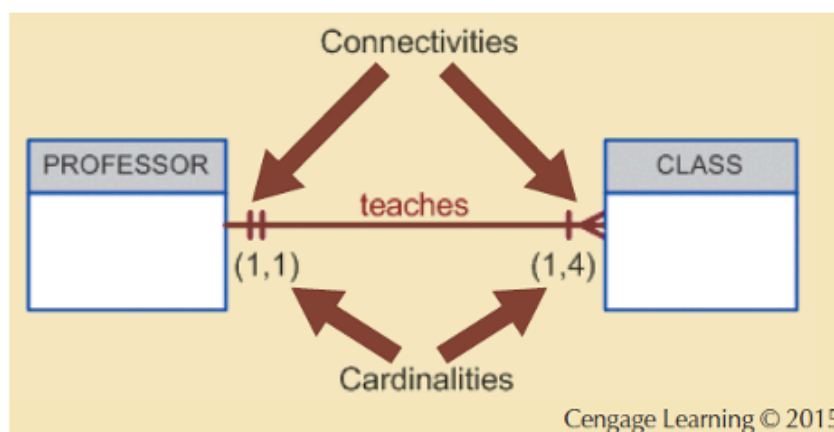
Relationships

- Association between entities that always operate in both directions
- **Participants:** Entities that participate in a relationship
- **Connectivity:** Describes the relationship classification
- **Cardinality:** Expresses the minimum and maximum number of entity occurrences associated with one occurrence of related entity

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Figure 4.7 - Connectivity and Cardinality in an ERD



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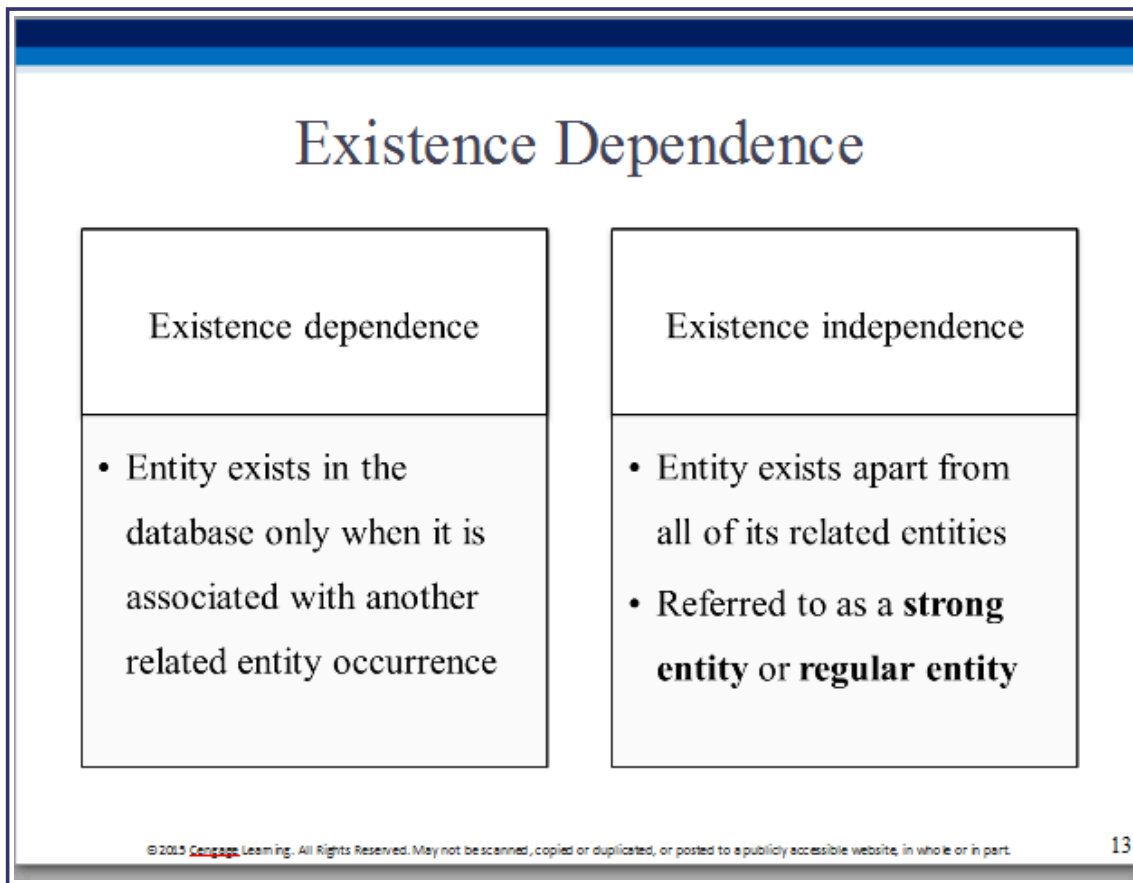
Connectivity: 1:1, 1:M or M:N (three diff ways by which two entities are related).

Cardinality: (min,max) for 1:1, 1:M or M:N (eg. 1:1 can have (1,0) as its cardinality, 1:M can have (0,4) as its cardinality). Sometimes, min is called 'modality' (and max is cardinality). The 'inside' symbols denotes min, and the outside ones, max.

Confusingly, the # rows in a table is ALSO called table's cardinality (and, # of columns is called the table's degree).

Also confusingly, 1:1, 1:M, M:N are called 'cardinality ratios'!

In/dependence ['can I exist apart from you?']



An entity B is "existent dependent" on another entity A, if, a row in B can only exist when its FK is NOT NULL, ie. a corresponding entry exists in A.

Eg. if A is EMPLOYEE and B is DEPENDENT, a dependent (eg. child) in B can only exist if there is a corresponding employee (eg. Dad) in A. But - this alone DOES NOT make 'B' a weak entity!

LOW: existence independence implies a strong entity; but, existence dependence (alone, ie. by itself) does NOT imply a weak entity (there needs to be one more condition, based on 'relationship strength', for it to become 'weak').

In other words, **we need to look at where the FK in the dependent entity is located**, to determine if the dependent entity is weak, or not-weak.

Relationship strength [weak vs strong]

Again, it's all about the FK [WHERE it goes], in the dependent entity!

Relationship Strength

Weak (non-identifying) relationship

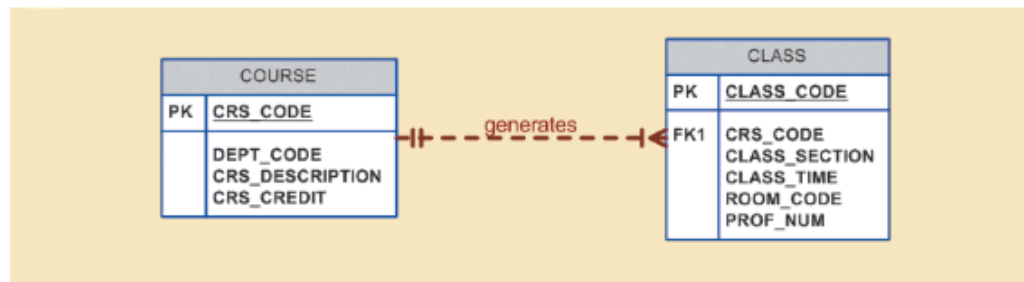
- Primary key of the related entity does not contain a primary key component of the parent entity

Strong (identifying) relationships

- Primary key of the related entity contains a primary key component of the parent entity

Not-weak, weak

Figure 4.8 - A Weak (Non-Identifying) Relationship between COURSE and CLASS



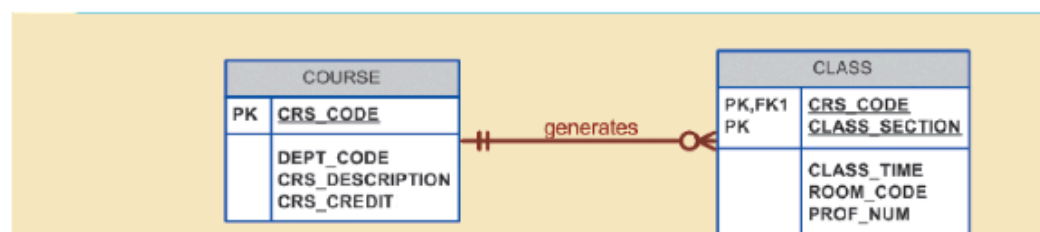
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In the above, CLASS is **not** a weak entity [is existent-dependent BUT has a weak relationship].

Figure 4.9 - A Strong (Identifying) Relationship between COURSE and CLASS



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CLASS above is now a weak entity (is existence dependent AND has a strong relationship). 'Strong' means 'shared PK'.

Weak entity [two conditions]

Weak Entity

- Conditions
 - Existence-dependent
 - Has a primary key that is partially or totally derived from parent entity in the relationship
- Database designer determines whether an entity is weak based on business rules

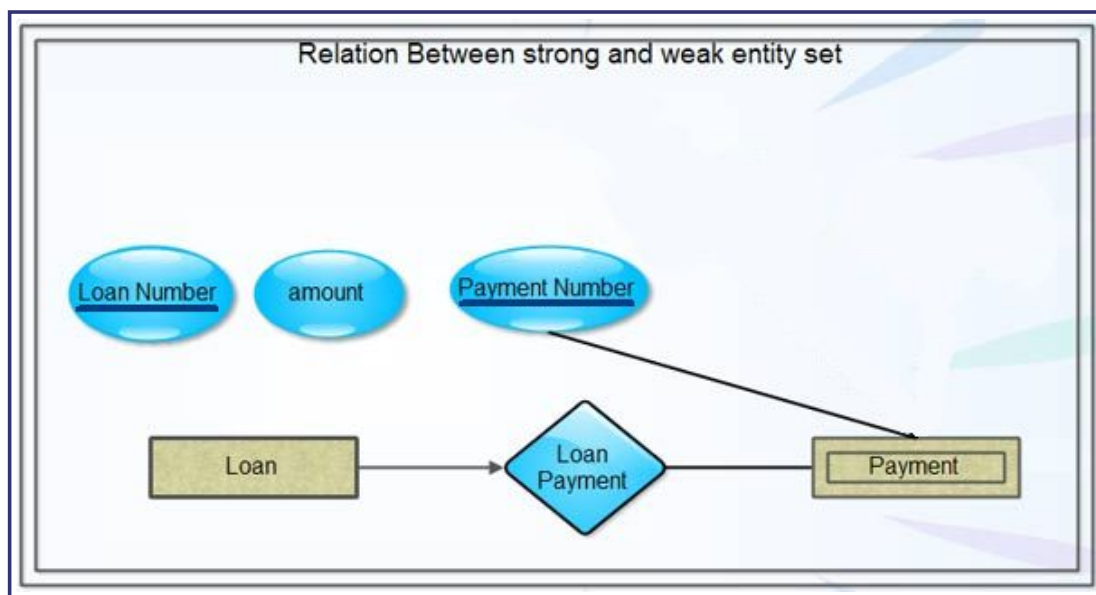
A weak entity needs to satisfy two conditions: existence dependence, strong (identifying/owning) relationship with a parent.

Note that a weak entity implies existence dependence, but existence dependence does not imply a weak entity!

Note too that a weak entity implies a strong ("owning" or "identifying") relationship.

Removing the controlling (owning) entity's key from a weak entity's PK will result in ****duplicates**** for remaining PK(s) - THAT is what makes it 'weak'.

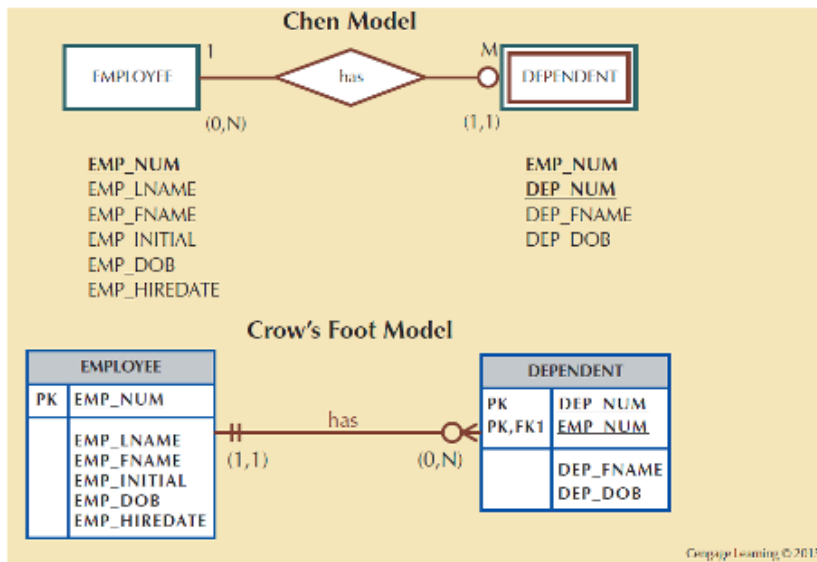
Weak entity - another example



Payment cannot exist independent of Loan, AND needs Loan's key to be part of its own key, so it (Payment) is a weak entity.

Weak entity - ERD and table depiction

Figure 4.10 - A Weak Entity in an ERD



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Figure 4.11 - A Weak Entity

Table name: EMPLOYEE **Database name: Ch04_ShortCo**

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Califante	Jeanine	J	12-Mar-64	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT

EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

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Relationship participation

Relationship Participation

Optional participation

- One entity occurrence does not require a corresponding entity occurrence in a particular relationship

Mandatory participation

- One entity occurrence requires a corresponding entity occurrence in a particular relationship

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Table 4.3 - Crow's Foot Symbols

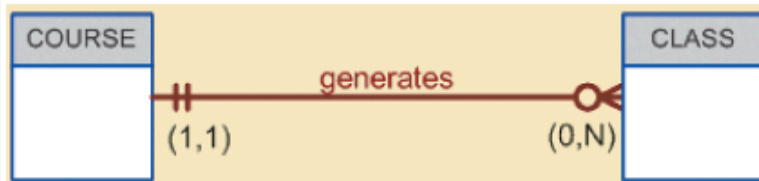
CROW'S FOOT SYMBOLS	CARDINALITY	COMMENT
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

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Figure 4.13 - CLASS is Optional to COURSE

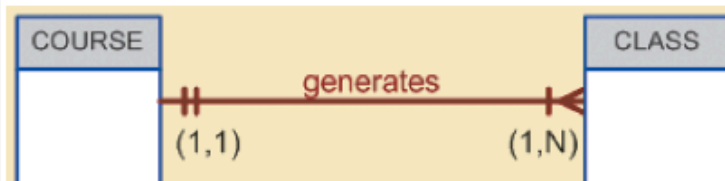


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Figure 4.14 - COURSE and CLASS in a Mandatory Relationship



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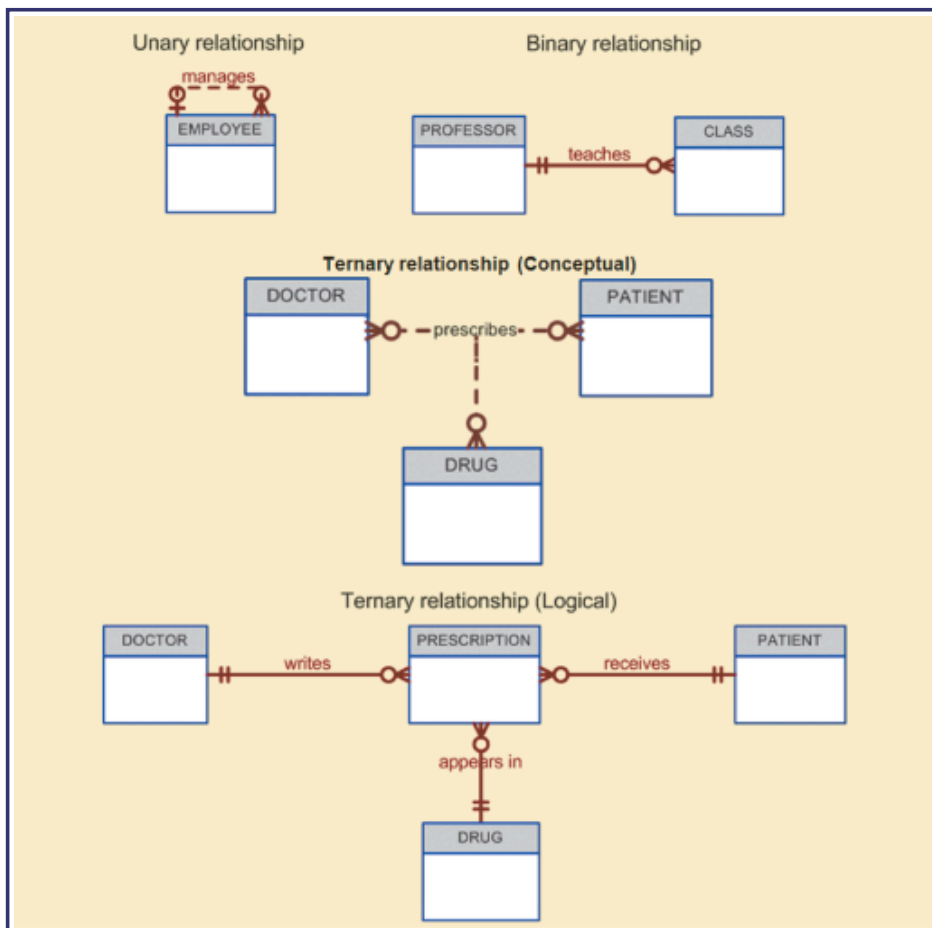
Relationship DEGREE

Relationship Degree

- Indicates the number of entities or participants associated with a relationship
- **Unary relationship**: Association is maintained within a single entity
 - **Recursive relationship**: Relationship exists between occurrences of the same entity set
- **Binary relationship**: Two entities are associated
- **Ternary relationship**: Three entities are associated

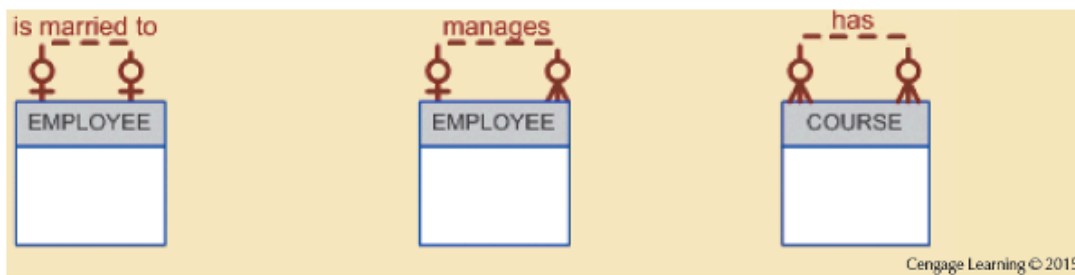
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Recursive:

Figure 4.17 - An ER Representation of Recursive Relationships



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Bridge entities

Associative Entities

- Also known as composite or bridge entities
- Used to represent an M:N relationship between two or more entities
- Is in a 1:M relationship with the parent entities
 - Composed of the primary key attributes of each parent entity
- May also contain additional attributes that play no role in connective process

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Figure 4.23 - Converting the M:N Relationship into Two 1:M Relationships

Table name: STUDENT

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Database name: Ch04_CollegeTry

Table name: ENROLL

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS

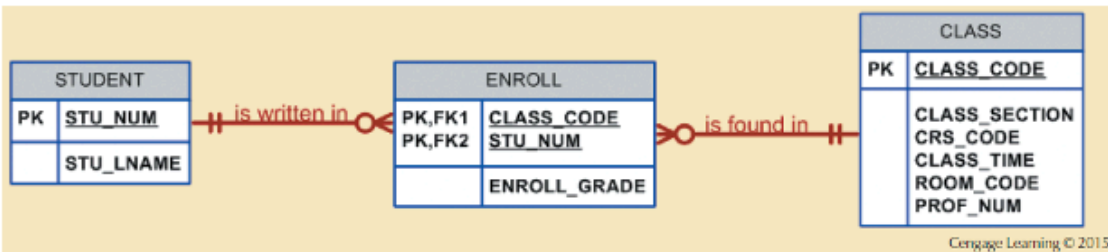
CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWTF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWTF 8:00-8:50 a.m.	KLR200	114

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Figure 4.25 - A Composite Entity in an ERD



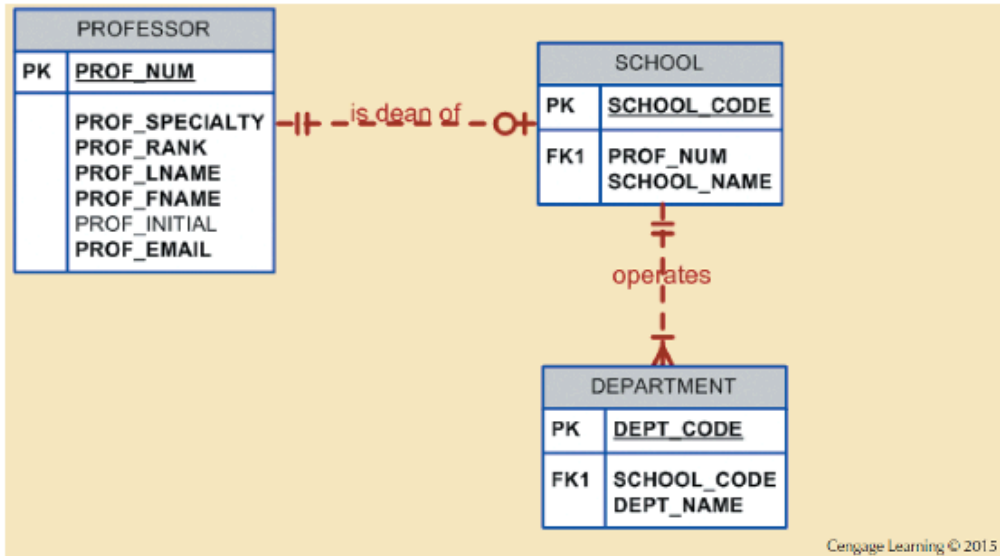
Putting together an ERD

Developing an ER Diagram

- Create a detailed narrative of the organization's description of operations
- Identify business rules based on the descriptions
- Identify main entities and relationships from the business rules
- Develop the initial ERD
- Identify the attributes and primary keys that adequately describe entities
- Revise and review ERD

ERD - piece by piece

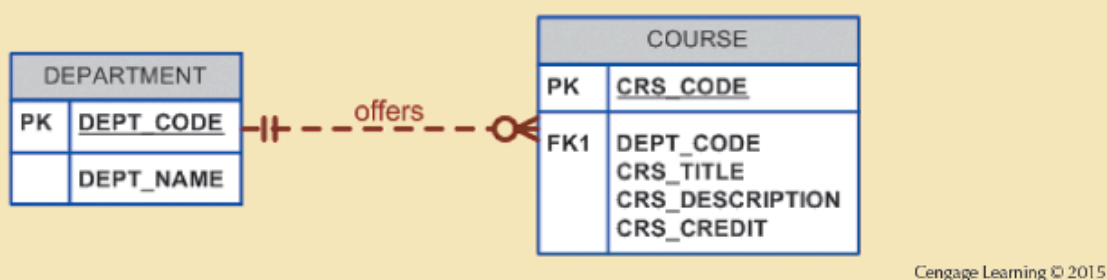
Figure 4.26 - The First Tiny College ERD Segment



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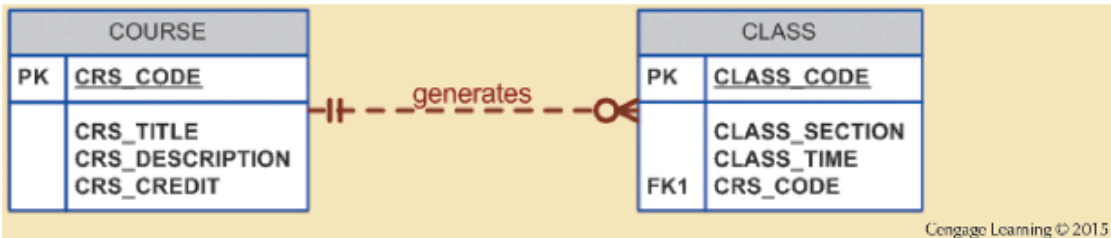
Figure 4.27 - The Second Tiny College ERD Segment



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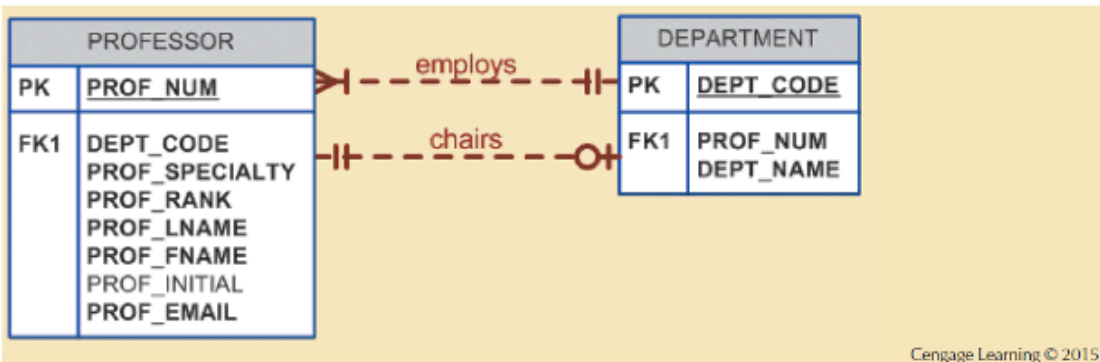
Figure 4.28 - The Third Tiny College ERD Segment



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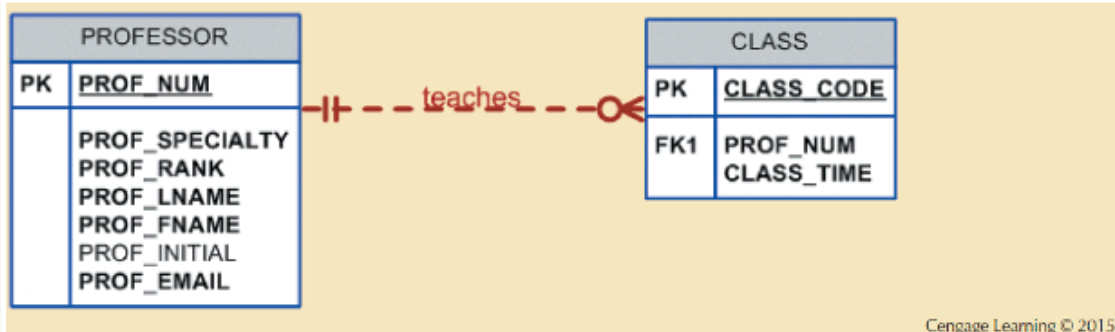
Figure 4.29 - The Fourth Tiny College ERD Segment



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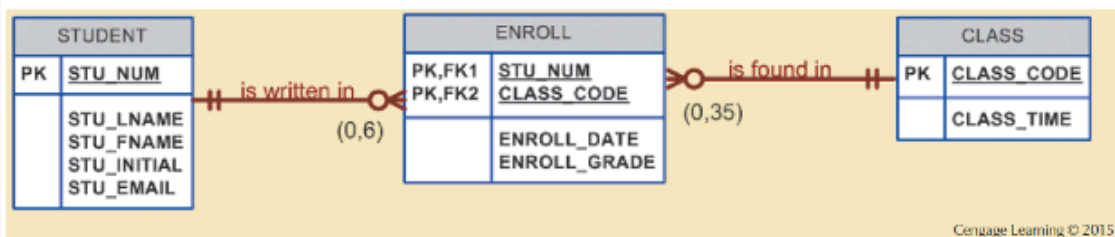
Figure 4.30 - The Fifth Tiny College ERD Segment



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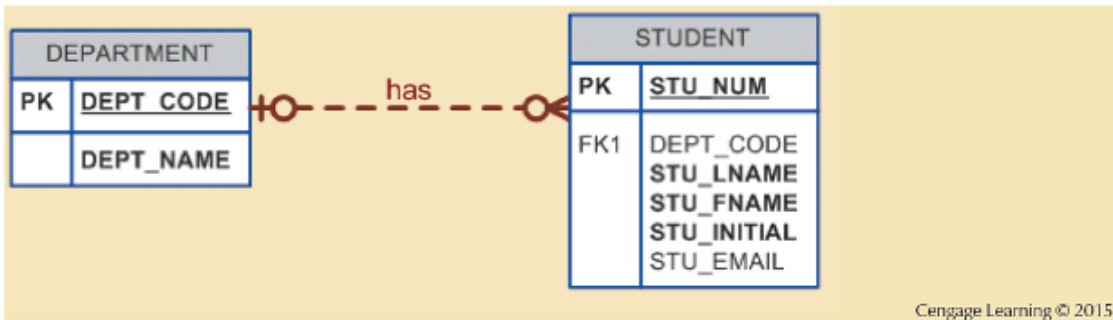
Figure 4.31 - The Sixth Tiny College ERD Segment



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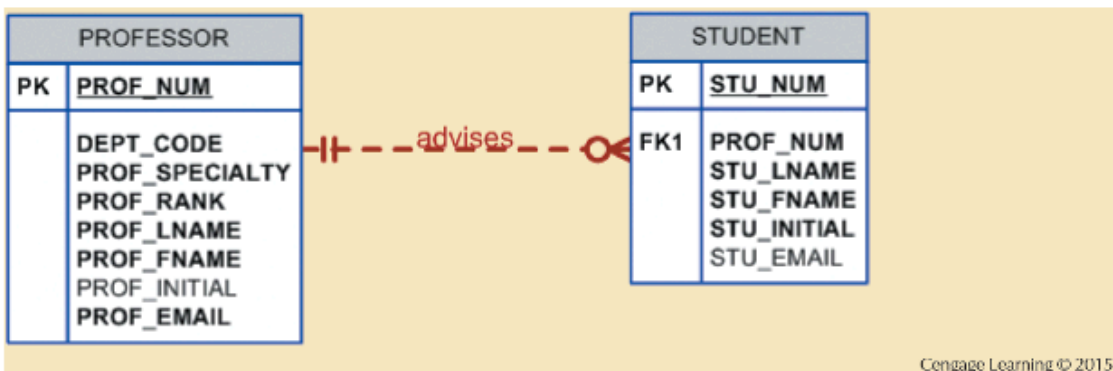
Figure 4.32 - The Seventh Tiny College ERD Segment



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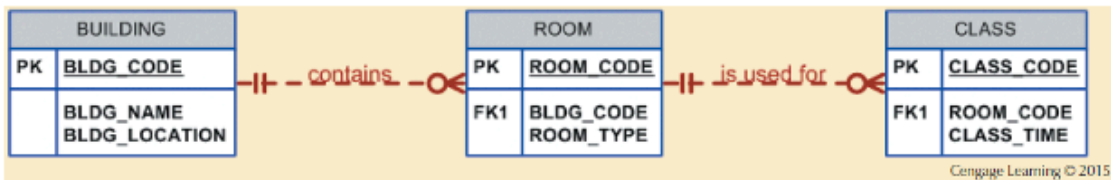
Figure 4.33 - The Eighth Tiny College ERD Segment



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Figure 4.34 - The Ninth Tiny College ERD Segment



The full schema

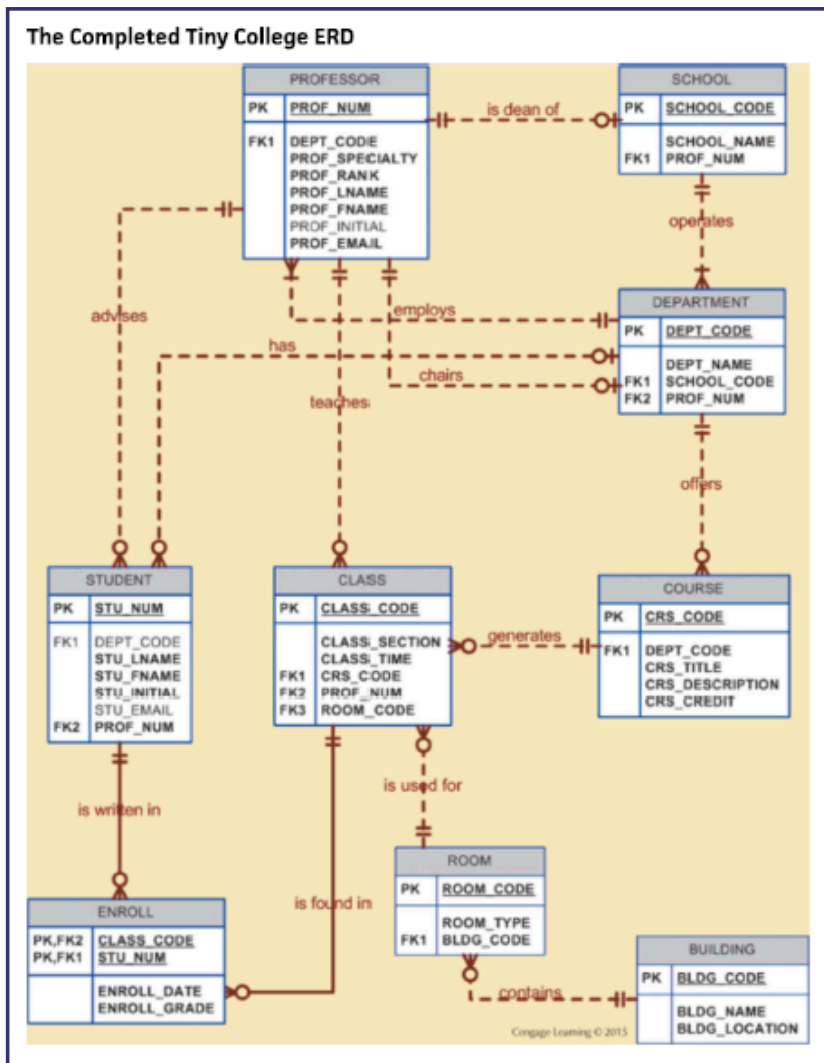
The full list ['BOM'] can be presented as a table, or as a diagram:

Table 4.4 - Components of the ERM

ENTITY	RELATIONSHIP	CONNECTIVITY	ENTITY
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS

Note: ENROLL is the composite entity that implements the M:N relationship "STUDENT enrolls in CLASS."

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"All together now!"